

NIKITA RAVI

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SUMMARY

Data & Analytics Engineer with 3+ years of experience designing end-to-end data platforms and dbt transformation layers across Snowflake, Redshift, and BigQuery. Expert in dimensional modeling, analytical SQL, and building governed semantic layers powering executive dashboards. Proven track record of translating raw, multi-source data into business-critical KPIs and scalable analytics systems across healthcare, public health, and B2B SaaS domains. Expertise in building production RAG, AI-native applications and using AI-assisted workflows.

EDUCATION

George Washington University — Master of Science, Data Analytics	May 2026
Dayananda Sagar College of Engineering — Bachelor of Engineering, Computer Science	Nov 2021

PROFESSIONAL EXPERIENCE

Zimmer Biomet , <i>Contractor Data Analyst (Remote)</i>	Jun 2025 – Present
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- Designed and deployed SQL-based transformation pipelines converting raw device telemetry and surgical event logs from SAP into procedure-level analytical datasets, enabling standardized reporting across 100+ regions
- Wrote advanced SQL and conditional aggregations to compute device-level metrics over rolling 90-day windows, and device-per-procedure ratios across 100+ regions, surfacing trends previously invisible in raw source data
- Parsed and normalized device data from FHIR endpoints and SAP extracts using Python, reconciling mismatched field formats and nested JSON structures across 5 source systems into a single governed schema for the commercial analytics team
- Implemented 8 device KPI definitions into the Power BI semantic model based on specifications from the analytics lead, ensuring consistent calculations across all downstream dashboards
- Built automated data validation checks within the refresh pipeline reducing analyst-reported data incidents by 40%
- Performed regression analysis on 12-month device usage and patient recovery data across 10K+ records using Python, isolating device-type and procedure-complexity variables that predicted post-surgical engagement decline
- Developed Power BI dashboards with drill-down views by territory, device type, and surgeon cohort, enabling commercial leadership to identify underperforming regions and prioritize field team resources
- Collaborated with clinical and commercial analysts to document metric definitions, transformation logic, and refresh schedules, reducing dashboard development cycles from 3 weeks to under 5 days

Data Science for Sustainable Development , <i>Data Engineer (Washington, DC)</i>	Aug 2024 – Dec 2025
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- Built cross-country health indicator rankings and year-over-year trend analysis with SQL across 194 nations by transforming raw WHO, World Bank, and UN data in Snowflake to identify statistical outliers
- Built dbt transformation layers (staging to intermediate to mart) standardizing 5 inconsistent API formats into a governed Snowflake schema with incremental refresh
- Centralized 15+ conflicting indicator definitions across 5 research teams into version-controlled dbt models with auto-generated documentation, establishing the first single source of truth
- Designed star-schema dimensional models in Snowflake with country, indicator, and time-period dimensions, enabling longitudinal cross-country analysis across 500+ health indicators
- Implemented automated dbt tests for nulls, uniqueness, referential integrity, and freshness across every model, catching 10+ silent data issues
- Developed country-level health scorecards in Tableau with self-serve filtering by region and income group, enabling research leads to independently identify outliers without engineering support
- Assisted in the development of a Retrieval-Augmented Generation system with Reciprocal Rank Fusion over 300+ SDG indicators using LlamaIndex and Qdrant, enabling researchers to query health and sustainability metrics in natural language and improving indicator discoverability by 30%

- Automated ingestion of 10K+ documents monthly by building AWS ETL pipelines (S3, Glue, Redshift) with OCR-powered extraction and anomaly detection, reducing processing errors by 40% and saving \$100K annually in manual review costs
- Built CI/CD pipelines for data workflows using Git, automating testing, validation, and deployment of ETL jobs and schema changes across AWS (Glue, Redshift), reducing deployment time and improving pipeline reliability
- Architected the document lifecycle tracking system using Neo4j graph database to model relationships between documents, processing stages, and client workflows enabling real-time lineage queries
- Developed the data layer powering a React workflow dashboard tracking document lifecycle status across 50+ B2B clients, reducing manual status checks by 25%
- Drove 22% increase in user engagement by partnering with Product and UX teams to analyze 50K+ activity logs using SQL and Python, presenting funnel analysis to leadership that improved B2B client retention by 12% and directly impacted enterprise account revenue for clients including IBM
- Designed KPI frameworks for 3+ enterprise clients defining engagement score, activation rate, and churn risk metrics, operationalized through automated daily SQL pipelines in Redshift and surfaced through Power BI dashboards used in quarterly business reviews
- Reduced downstream reporting errors by 35% by implementing schema enforcement, null checks, and anomaly detection at each pipeline stage, with reconciliation queries tracing root causes across Lambda, Glue, and Redshift layers

PROJECTS

Pharma Evidence RAG System | pharma-evidence-rag.onrender.com

- production RAG system with FastAPI, Qdrant, BM25 and hybrid retrieval ingesting from PubMed, FDA, and ClinicalTrials.gov, improving retrieval accuracy by 22% through cross-encoder reranking built for GLP-1 and women related drugs.

NYT Analytics Engineering Pipeline | github.com/nikita-ravi/nyt-analytics-engineering-pipeline

- Designed end-to-end analytics engineering platform using dbt, Snowflake, and Airflow with dbt, automated testing, and CI/CD deployment

Healthcare Claims Analytics | <https://github.com/nikita-ravi/claims-anamoly-platform>

- Analyzed 1.4M+ Medicare/Medicaid claims across 37,780 patients and 2,675 providers using SQL and Python, identifying \$2.3M in billing anomalies through statistical outlier analysis

Cross-Lingual Depression Detection | <https://github.com/nikita-ravi/Cross-Lingual-Depression-Detection>

- Investigated cross-lingual transfer learning for mental health screening by fine-tuning RoBERTa on 40K English Reddit posts and evaluating on Arabic and Russian datasets, achieving 15% accuracy improvement over baseline through English-to-Arabic knowledge transfer

TECHNICAL SKILLS

Languages: SQL, Python, R, JavaScript

Pipeline & Orchestration: Airflow, dbt, Spark, PySpark, Kafka, AWS Glue, Lambda, ETL/ELT

Cloud & Warehouse: Snowflake, BigQuery, AWS, GCP

Visualization & BI: Tableau, Power BI, Streamlit, Looker

Statistics & Analytics: EDA, A/B Testing, Hypothesis Testing, Time Series Forecasting, Regression, Feature Engineering, Anomaly Detection, Statistical Validation, Ad Hoc Analysis

AI/ ML: Random Forest, XGBoost, Logistic Regression, Clustering, K-Means, DBSCAN, Scikit-learn, TensorFlow, NLP

Tools: Git, Docker, Linux, CI/CD, Railway, Render, MLOps

LLMs & GenAI: RAG Pipelines, LangChain, Llamaindex, Hugging Face, Qdrant, FAISS, Prompt Engineering, Looker, VertexAI, Claude Code, Antigravity